

Delhi Public School

Subject: Physics | Grade: 10th | Class: ____

Time: 2 Hours

Maximum Marks: 11

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Answer all questions. Each question carries 1 mark. Choose the correct option.

Q1. Which of the following materials cannot be used to make a lens? [Easy] [1]

- a. Water
- b. Glass
- c. Plastic
- d. Clay

Q2. A student traces the path of a ray of light passing through a rectangular glass slab for four different values of angle of incidence. Which of the following correctly represents the relation between the angle of incidence (i), angle of refraction (r), and angle of emergence (e)? [Moderate] [1]

- a. $i > r = e$
- b. $i = e > r$
- c. $i < r < e$
- d. $i = r < e$

Q3. An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be: [Moderate] [1]

- a. 100 W
- b. 75 W
- c. 50 W
- d. 25 W

Q4. A cylindrical conductor of length l and uniform area of cross-section A has resistance R . Another conductor of length $2l$ and resistance R of the same material has area of cross-section: [Hard] [1]

- a. $A/2$
- b. $3A/2$
- c. $2A$
- d. $3A$

Q5. At the time of short circuit, the current in the circuit: [Easy] [1]

- a. reduces substantially
- b. does not change
- c. increases heavily
- d. varies continuously

Section B

Answer the following short answer questions. Each question carries 2 marks.

Q6. A concave mirror produces a three times magnified real image of an object placed at 10 cm in front of it. Find the location of the image. [Moderate] [2]

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Q7. Why does the Sun appear reddish early in the morning? Explain the physical phenomenon involved. **[2]**

[Moderate]

Q8. Two resistors with resistances R_1 and R_2 (where $R_1 > R_2$) are connected first in series, and then in parallel. In which of these cases will the equivalent resistance be (i) higher than R_1 , and (ii) lower than R_2 ? Justify your answer mathematically. **[2]**

[Hard]

— End of Question Paper —